Quantifying Human Expressivity in Keyboard Performance: A Comparative Analysis of Dynamic and Temporal Variance in the MAESTRO Dataset

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Abstract

Musical expression exists on a wide spectrum, and recent studies have put significant effort into understanding the variety of expressivity that reaches and resonates with the human mind. This study presents a computational framework for quantifying musical expressivity by comparing human piano performances against a synthesized mechanical baseline. Using a stratified sample of 30 performances from the 2018 MAESTRO dataset, features were extracted from MIDI note-on events to analyze two primary dimensions of expression: Dynamic Range (via Velocity Standard Deviation) and Temporal Rubato (via Inter-Onset Interval jitter).

The results establish a clear statistical divergence between human and mechanical models. Human performances ranged around a dynamic signature of $\sigma \approx 20$ MIDI velocity units, while the mechanical baseline exhibits zero variance. Particularly, temporal analysis revealed that timing jitter varies more significantly across top-tier performers than dynamic range, which suggests that rubato serves as a primary vehicle for individual artistic identity. By mapping these features into a two-dimensional feature space, this research provides a metric for "humanity" in performance—quantifying the subtle deviations that allow a performer to interpret a rigid musical score through expressive wit. By bridging the gap between raw MIDI data and high-level artistic intent, this framework serves as a foundation for more empathetic musical technologies that respect the distinct wit of the human performer.

1 Introduction

With the emerging advancements of technology, many music content generated by AI have been circulating, often blurring the lines between synthetic and human-led artistry. While acknowledging that these computational systems can replicate the structural accuracy of a musical score, they frequently struggle to capture the level of "human feel" that defines a virtuosic performance. The central problem addressed in this study is the lack of a standardized, quantitative metric to distinguish the subtle dynamic and temporal deviations of a human pianist from the rigid, literal interpretations of a computer-generated baseline. This research hypothesizes that human expressivity is not a random occurrence but a structured, high-entropy signature that can be isolated by measuring the standard deviation of velocity and timing jitter. By comparing human data from the MAESTRO dataset against a de-humanized mechanical control, this study aims to prove that "humanity" in music is a statistically significant departure from a perfectly quantized grid.

The study aims to establish a two-dimensional feature space that maps these micro-deviations, providing a clear visual and mathematical boundary between artistic intent and mechanical playback. This work offers a potential benchmark for developers in the AI music space to refine the "emotional intelligence" of their algorithms, ensuring that future technologies respect the traditional "contract" of musical performance while incorporating the nuanced wit of the human performer. By quantifying the dual-focus of dynamics and timing, this research underscores that the nuanced wit of a human performer, the spontaneous, non-linear choices that breathe life into music, remains a uniquely human signature. Recognizing these distinctions is a critical step toward ensuring that future musical technologies respect the depth of human artistic intent rather than merely approximating its surface.

2 Related Work

The study of musical expressivity sits at the intersection of mathematical precision and human emotion. At its core, expressivity is the "wit" that lives within the "walls" of a musical score; it is the intentional deviation from a literal, mechanical interpretation that allows a performer to communicate tension, release, and narrative. In the field of Music Information Retrieval (MIR), quantifying this elusive "human feel" has evolved from simple empirical observations to complex computational models. As AI-generated music continues to circulate, understanding these fundamental deviations becomes essential for distinguishing between a performance

that merely follows a series of notes and one that authentically connects with a listener. To provide a rigorous foundation for this analysis, this study draws upon pivotal areas of modern research.

To capture music expressivity that is capable of capturing micro-nuances of a master pianist, Hawthorne et al. (2019) introduced the MAESTRO dataset, which remains the gold standard for aligned MIDI and audio performance data. Their work emphasizes that for a machine to understand expressivity, it must first be trained on "virtuosic" data where every millisecond of timing and every unit of velocity reflects a deliberate artistic choice. This dataset provides the high-resolution "ground truth" necessary to observe the dynamic and temporal signatures that define human artistry, moving beyond the limitations of standard, quantized MIDI files.

Building upon this data, recent research has shifted toward understanding how these technical deviations are perceived as "expressive" by a human audience. Chowdhury and Widmer (2024) explore this through the lens of mid-level perceptual features, arguing that expressivity is not just a collection of random errors but a coherent set of descriptors—such as "articulation" and "rhythmic clarity"—that computers can be trained to recognize. Their work justifies the use of statistical variance (Standard Deviation) as a proxy for expressivity, as these low-level fluctuations in velocity and timing are the fundamental building blocks that constitute the "perceptual warmth" of a live performance.

A recent quantitative analysis of piano performance also supports the use of velocity standard deviation as a proxy for expressivity. Kim et al. (2021) conducted a large-scale study comparing experts and amateurs, demonstrating that "handcrafted features" representing dynamic attributes, specifically the mean and standard deviation of MIDI velocity, are significant factors in determining performance proficiency. The statistical findings of this study showed that expert performers were able to play a more controlled and “smoother change” in dynamics along melodic lines compared to amateurs. Moreover, their use of Velocity SD as a feature in principal component analysis (PCA) and support vector machine (SVM) classification confirmed that these measures of dynamic variance can effectively identify the “human feel” and technical mastery associated with high-level artistry.

The same study also utilized IOI-based features to examine temporal stability and rhythmic control, finding that the ability to intentionally manipulate timing without losing structural coherence is a hallmark of professional proficiency. Their analysis of note-based features reveal that expert performers use micro-deviations in IOI to emphasize melodic importance, a technical behavior that distinguishes human performance from the rigid periodicity of quantized MIDI.

Furthermore, Schreiber et al. (2020) applied Kernel Density Estimation in their study suggesting that musical time is rarely a static constant, but it is a distribution of events seen as a coherent pulse. By highlighting the limitations of single-value metrics in tempo and timing analysis, their work justifies the use of KDE to visualize the continuous probability density of Inter-Onset Intervals (IOIs). Through this approach, it treats tempo as a “probabilistic signature” rather than a mechanical fixed point.

By linking these perspectives, it becomes clear that expressivity is a quantifiable phenomenon rooted in structured variance. While the aforementioned studies utilize the features for classification, there remains a need to establish a direct, comparative baseline against a purely mechanical model.

3 Methodology

3.1 Data

The methodology for quantifying keyboard expressivity began with the selection and preprocessing of data from the MAESTRO (MIDI and Audio Edited for Synchronous Tracks and Organization) Dataset (v3.0.0). To ensure consistency in sensor calibration and instrument response across the sample, a stratified sampling approach was used to select a subset of 30 MIDI performances from the 2018 competition year. Feature extraction and data management were executed within a Jupyter environment managed by Anaconda, utilizing the pretty_midi and pandas Python libraries to ensure a reproducible computational pipeline.

3.2 Approach

During the feature extraction phase, only the primary piano track (Instrument 0) of each MIDI file was

processed to isolate the performer's input. For every note event, three raw parameters were extracted: Velocity (v), representing the force of the key press on a scale of $0 - 127$; Onset Time (t), the precise start time of the note in seconds; and Duration (d), the time elapsed between the note-on and note-off events. Following extraction, a Synthetic Mechanical Baseline was generated for every performance to serve as a "de-humanized" control group. This process involved dynamic flattening, where all velocities were set to a constant median value of 64, and temporal quantization, where onset times were snapped to a rigid $0.125s$ grid to remove all rhythmic micro-deviations.

The study then defined two primary computational metrics to quantify the "human feel" of the performances. Dynamic Range σ_v was calculated as the standard deviation of note velocities to measure the performer's breadth of volume. Simultaneously, Timing Jitter (σ_{IOI}) was calculated using Inter-Onset Intervals (IOI), defined as $IOI = t_n - t_{n-1}$ for unique onset times. The standard deviation of these intervals served as a mathematical representation of Rubato, or expressive timing. Finally, the data was synthesized into two visual formats: Kernel Density Estimation (KDE) plots to visualize the distribution of expressive intent against the static mechanical limit, and 2D Feature Space Mapping, which plotted σ_v against σ_{IOI} to identify distinct clusters of expressive artistry.

4 Results and Discussion

The comparative analysis between human performances and the synthesized mechanical baseline revealed a significant "expressivity gap" across both dynamic and temporal dimensions. The mechanical baseline yielded a zero-variance profile with a constant MIDI velocity of 64 and a perfectly uniform Inter-Onset Interval (IOI) of 125ms. In contrast, the human performances from the 2018 MAESTRO dataset demonstrated a high degree of entropy, suggesting that virtuosic playing is defined by intentional deviations from a rigid musical grid.

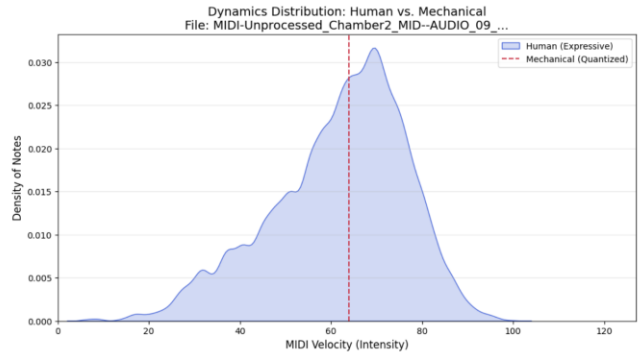

Figure 1, The Velocity Distribution of Human vs Mechanical Performance

The Kernel Density Estimation (KDE) analysis of MIDI velocities (Figure 1) highlights the range of human dynamic intent. While the mechanical baseline is represented by a single, infinite spike at the median, the human distribution presents a broad, multi-modal curve. Notably, the human data shows a significant density in the lower velocity ranges ($v < 40$), indicating that the "subtle touch" or *pianissimo* is a primary feature of expressive performance that is lost in standard MIDI quantization. The mean standard deviation for the human sample ($\sigma_v \approx 18 - 20$) suggests a consistent "dynamic signature" that performers maintain to create musical contrast and emotional contour.

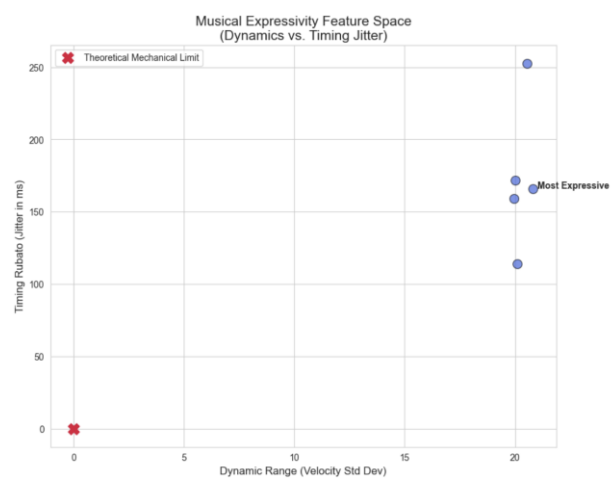

Figure 2, The 2D Feature Space Mapping of Dynamic vs Timing Jitter

When these dynamics were mapped against temporal jitter in a two-dimensional feature space (Figure 2), a distinct clustering effect was observed. The human performances occupied a specific region of the feature space, significantly removed from the theoretical

mechanical limit at the origin (0,0). While the dynamic range (σ_v) remained relatively stable across the top-tier performances, the timing jitter (σ_{IOI}) showed a much wider variance, ranging from approximately 110ms to 250ms. This suggests that while dynamic breadth is a baseline requirement for virtuosic performance, the manipulation of time—or rubato—is a more highly individualized expressive choice. The clear separation between the human cluster and the mechanical limit provides a mathematical "humanity metric" that can be used to assess the expressive quality of computer-generated music.

5 Conclusion and Recommendations

This study confirms that musical expressivity is a quantifiable phenomenon that can be effectively modeled through the lens of dynamic and temporal variance. It also successfully isolates the distinctive parameters of “human feel” through comparing the rigid, zero-variance nature of a synthesized mechanical baseline with the high-entropy signatures of virtuosic human performance. The data suggests that while human performers operate within the shared "walls" of a musical score, their identity is defined by a consistent mastery of the pianissimo-to-fortissimo spectrum and an individualized approach to rhythmic rubato. The observed clustering in the two-dimensional feature space supports that expressivity is not a random deviation from a grid, but a structured, intentional expansion of the musical sheet that translates technical precision into emotional connection.

Furthermore, expanding the dataset to include diverse genres beyond the classical performances in the MAESTRO set would allow for a cross-cultural comparison of expressive signatures. Such an expansion would determine if the observed dynamic clusters are universal to human musicality or specific to the Western pedagogical traditions represented in the current sample. By broadening the scope of the data, researchers can identify whether different genres possess unique "expressivity fingerprints" that require distinct statistical baselines.

Finally, these findings suggest that automated assessment tools should prioritize the "spread" of a performance rather than its proximity to a perfect temporal grid. Developing real-time visualization tools based on the feature space mapping presented in this study could provide piano students with immediate, intuitive feedback on their expressive range. By bridging the gap between raw MIDI data and high-level artistic intent, this framework serves as a foundation for more empathetic musical technologies that respect the nuanced wit of the human performer. The statistical bounds identified in this research can serve as a validation metric to ensure that machine-generated music preserves the essential "human feel" required for deep listener connection.

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